Investigation of the Learning Process Through Game Development

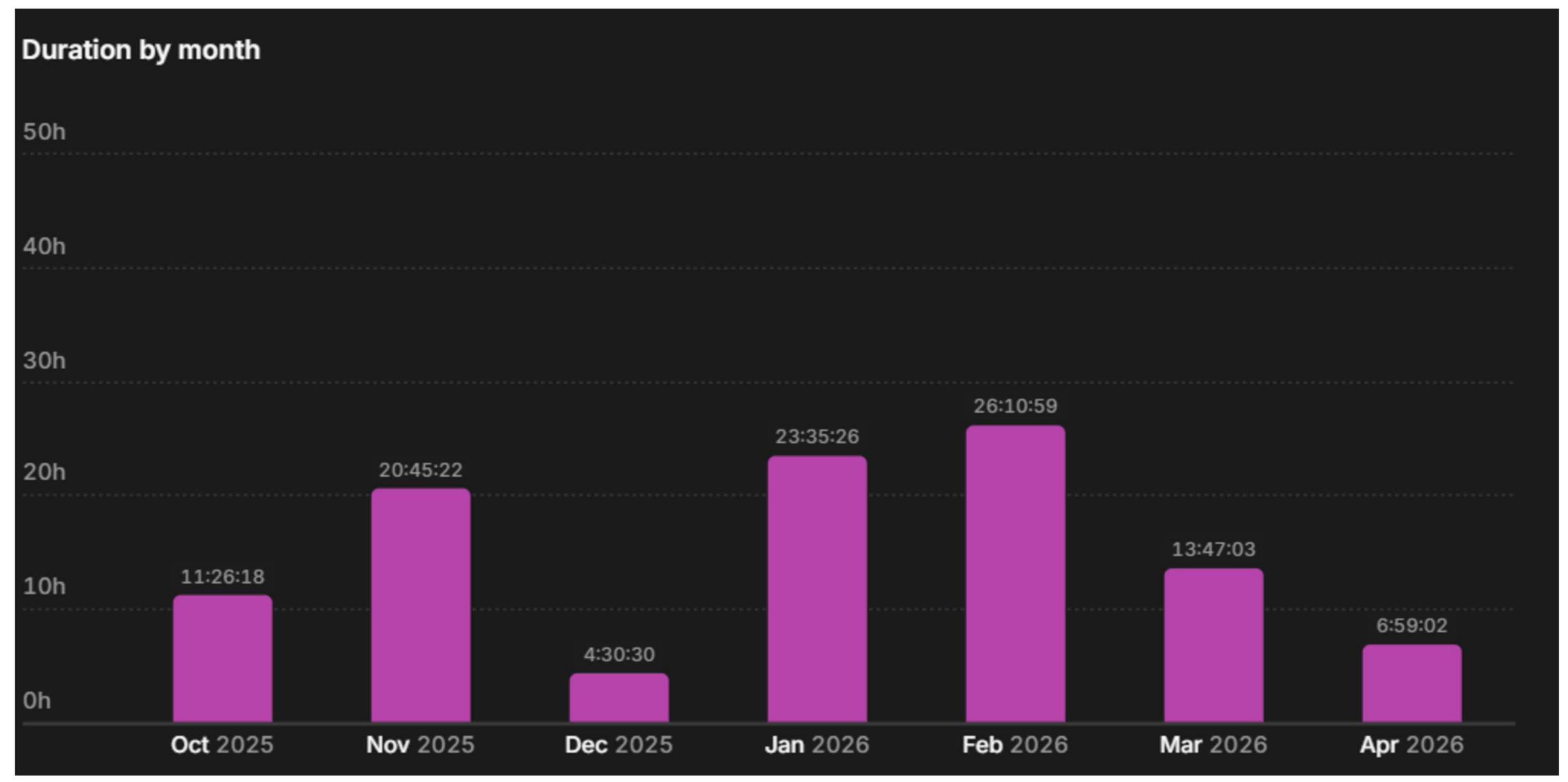
Background

- Learning is hard and many different factors can negatively impact it.
- A lot of research about specific skills and techniques in isolation but not how to navigate learning including the confounding variables of life.
- **Goal**: Learning about learning through making a video game.
- Game development was chosen to encompass many different fields and encounter new situations.

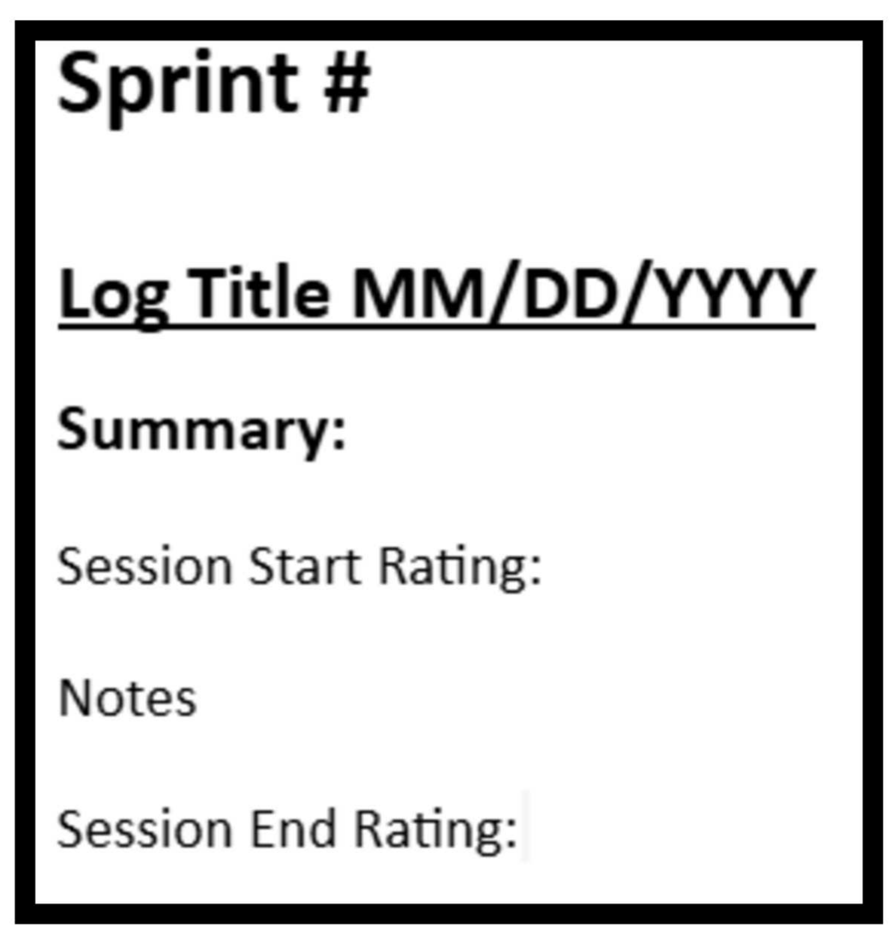
Methods

- Learning journal to track each day's work and findings
- Session lengths are also tracked to see variations over time and between logs
- Sprint Work Organization
  - ~2 weeks of time
  - Plan at start
  - Retrospective at end
- Utilized free resources
  - Godot 4.6 - Game Engine
  - Toggl Track - Time Tracker
  - Lucid Chart - Flowchart Creator
  - Blender 5.0 - 3D Modelling
  - YouTube & Online Forums - Game Development Skills

Results



Time Tracked from October 24, 2025 - April 23, 2026



Sprint #

Log Title MM/DD/YYYY

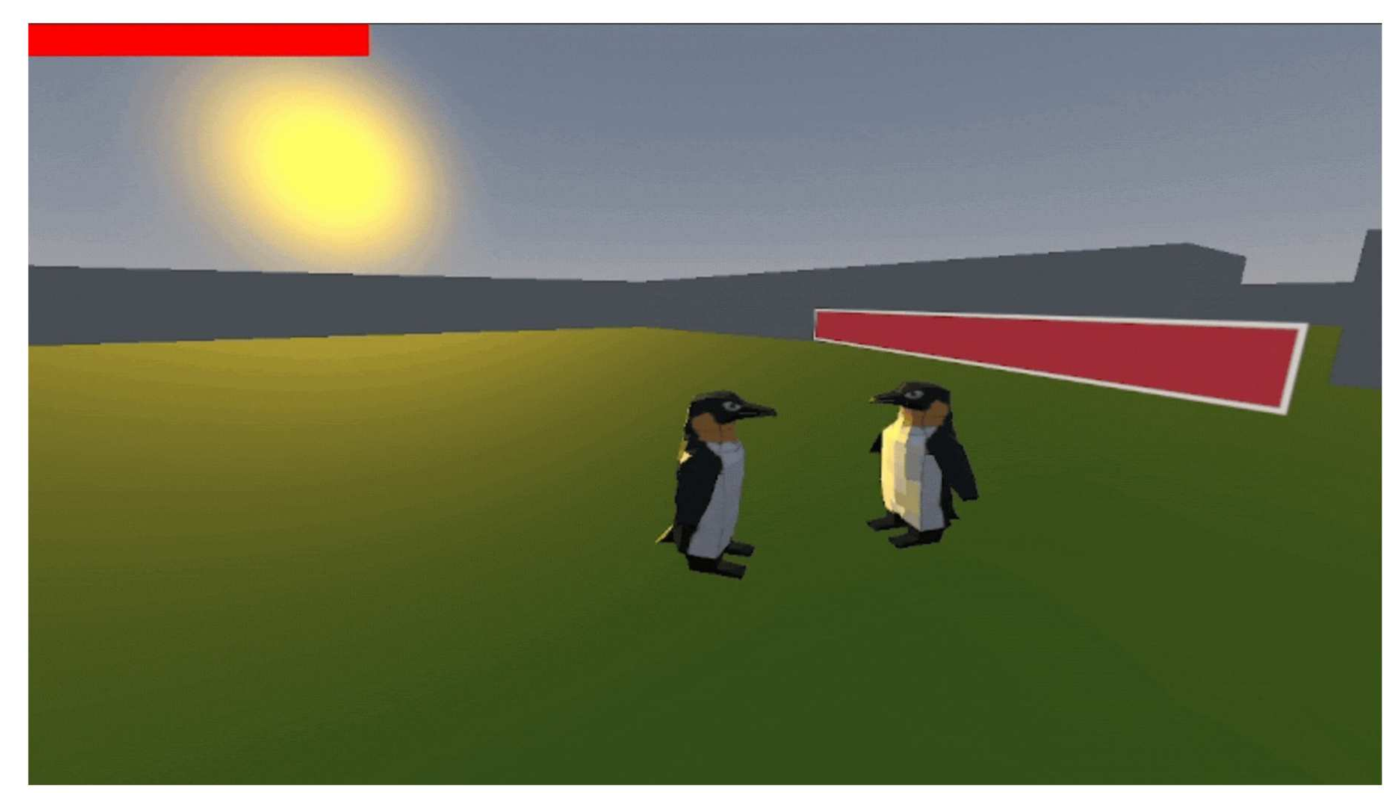
Summary:

Session Start Rating:

Notes

Session End Rating:

Notebook log Template



In-game screenshot of player and enemy

- Two types of logs
  - Reflection-heavy
  - Work-heavy
- Very cliché-like or common sense ideas
  - Be proud of the work being put in
  - Derive enjoyment from the process of work and not just the fruits of labor
- Without structure or specific goals, time put in fluctuates

Discussion / Conclusions

- Control external environment to create a more suitable work environment
- Work with others to receive validation and feedback on work done
- Hardest part is consistent practice
- Perfectionism caused biggest issues
  - Dissatisfaction with work
  - Disoriented sense of scale
- Internal and external conditions work hand-in-hand
  - Less doomscrolling

References

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